

A SYSTEMATIC REPLICATION OF THE PRESCRIPTIVE TREATMENT OF SCHOOL REFUSAL BEHAVIOR IN A SINGLE SUBJECT

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Summary — We employed a multiple-baseline design to test the efficacy of a recently developed approach for reducing school refusal behavior. This approach uses a systematic method for conducting a functional analysis for the target behaviors and suggests specific intervention strategies based upon this analysis. To maximize external validity, the intervention was tested using a systematic replication strategy, whereby only the major conceptual elements of the intervention were retained from previous applications. The subject was a 10 year old girl who attended school with great difficulty and resistance and who was assigned DSM-III-R diagnoses of separation anxiety disorder and social phobia prior to treatment. Following 8 weeks of treatment, she showed marked reductions in problem behavior and no longer met criteria for any anxiety disorder diagnoses. The results highlight the importance of the use of functional analytic strategies to effect behavior change. Copyright © 1996 Elsevier Science Ltd

School refusal is a broad term used to describe the childhood syndrome characterized by unwillingness to attend school and recurrent absenteeism. Perhaps its most comprehensive definition comes from Pilkington and Piersel, 1991 (p. 291): “the hallmark symptom...is refusal to attend school or remain in school once there, despite pressure, punishment, and/or recrimination from parents, teachers and other school authorities”. The term “school refusal behavior” need not include actual absenteeism, but can refer also to the syndrome of associated behaviors that interfere with the child’s family, peer relations, or academic functioning. These commonly include features such as recurrent somatic complaints, chronic visits to the school nurse, crying before or during school, tantrums, aggression, or truancy. Until recently, there has been difficulty arriving at a clear formulation of the syndrome (Kearney, in press), and

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therefore, traditional approaches to treatment have been quite heterogeneous and lacked conceptual direction.

With regard to internalizing syndromes, school refusal behavior has historically been conceptualized in a number of different ways. For example, several attempts have been made to understand school refusal as a symptom of DSM-III and DSM-III-R anxiety or affective disorders (Bernstein & Garfinkel, 1986; Last & Francis, 1988; Last, Francis, Hersen, Kazdin, & Strauss, 1987; Last & Strauss, 1990; Ollendick & Mayer, 1984). Further, there is question as to whether the phenomenon is principally related to simple phobia (e.g., excessive fear of teachers), social phobia, separation anxiety, agoraphobia (e.g., afraid of feeling sick in school and having no way out), or an anxiety disorder not otherwise specified (e.g., test anxiety unrelated to social concerns). This diagnostic uncertainty has served to complicate school refusal nosology; for example, some researchers continue to use the terms "school phobia" and "school refusal" interchangeably, considering them to be more or less conceptual equivalents (cf. Burke & Silverman, 1987; Kearney, in press).

Current research on school refusal behavior acknowledges the prevailing conceptual inadequacy and speaks to the possibility of alternative nosological approaches (e.g., Kearney, 1995a; Pilkington & Piersel, 1991). One approach stems from basic behavioral principles of standardized functional assessment (Kearney & Silverman, 1990). This approach draws from the work of Durand (1990), who has focused on modifying challenging behaviors in children with developmental disabilities and has developed a programmatic functional assessment strategy for problem behaviors in children. In the context of treating school refusal behavior, this method has been termed the "prescriptive approach"* (Burke & Silverman, 1987; Kearney & Silverman, 1990). It attempts to identify functional components of the school refusal behavior, pointing to variables that may maintain the behavior. It most notably departs from a diagnostic approach in that school refusal behavior is classified based on function rather than topography. Although much of behavior therapy has always diverged from an emphasis on diagnosis (Nelson, 1988), the present approach is nevertheless unique in that, in contrast to traditional functional analysis, the assessment is mainly questionnaire-driven and therefore allows a rapid and unconstrained case formulation.

Kearney and Silverman (1990) identified four functional aspects of school refusal behavior: (1) specific fearfulness or general over-anxiousness; (2) escape from aversive social stimuli; (3) attention-getting or separation anxious behavior; and (4) tangible reinforcement. Functional assessment of school refusal should suggest one or a combination of these categories and have clear implications for treatment.

The purpose of the present study was to empirically evaluate prescriptive treatment of school refusal behavior. Although the prescriptive approach is theoretically compelling and has preliminary support for its efficacy (Kearney & Silverman, 1990), no systematic replication has been conducted. Sidman (1960) highlights the critical difference between direct and systematic replication in that the former represents identical duplication of the experimental design

*The term "prescriptive treatment" has been used elsewhere in the literature to refer to empirically based interventions differentially applied to specific disorders or their subtypes (e.g., Hersen & Ammerman, 1994). For the purposes of this manuscript, the term prescriptive treatment refers to the somewhat more general usage as articulated by Burke and Silverman (1987) to denote empirically-based interventions differentially applied to specific behavior problems or syndromes as an outcome of functional analysis.

whereas the latter allows several conditions to vary (cf. Kazdin, 1993). In the interest of evaluating the external validity of the prescriptive approach, treatment was conducted using the basic conceptual ingredients of functional assessment and treatment as suggested by Kearney and Silverman (1990). No attempt was made to duplicate factors such as length of treatment, specific session content, or session length; and no written protocol was used. In such a manner, this study attempted to strengthen the external validity of the prescriptive approach and at the same time to illustrate some of the challenges and critical features of a successful prescriptive intervention.

Method

Subject Description

The subject was a 10 year, 2 month old girl in the fourth grade, referred for assessment due to difficulties with school refusal behavior. She had no previous treatment history and was not on medication at the time of assessment or treatment. The initial assessment entailed the administration of the Anxiety Disorders Interview Schedule, Child and Parent Versions (ADIS-C/P; Silverman, 1991). These are separate structured clinical interviews focusing on the diagnosis of DSM-III-R anxiety disorders and their accompanying mood and behavioral disturbances. These instruments have demonstrated adequate diagnostic reliability for childhood anxiety disorders (Silverman & Eisen, 1992). Based on the information in the initial diagnostic interviews, the child was assigned a principal DSM-III-R diagnosis of Separation Anxiety Disorder and an additional diagnosis of Social Phobia.

According to the parents, the child had a history of school refusal behavior since the first grade, and it had been a daily struggle to get her to attend school. These problems abated during the second and third grades and reappeared at the start of fourth grade. Typically, school refusal behavior intensified each year in December and even more in January, following the winter holidays. According to the parents, the child cried daily about having to go to school and exhibited poor appetite and restless sleep. Despite these difficulties, however, the child had no extended periods of absenteeism, but rather, she endured school with marked distress.

At intake, the child reported that she often worried about her parents' well-being (especially her mother's) when she was separated from them. In particular, she worried that her parents could be hurt in some kind of accident. She was also concerned that something might happen to her while she was away from her parents. Accordingly, she usually felt sick or nauseous when it was time to leave home to go to school. During separation, she experienced moderate to strong levels of nausea, hot flushes, palpitations, breathlessness, derealization, lightheadedness, and a fear of dying. She claimed that these feelings had been present for as long as she could remember.

In addition to these fears of separation, the child experienced persistent apprehension about doing or saying something to embarrass herself in social situations. She worried that classmates would negatively evaluate her or make fun of her. Situations she reported as anxiety provoking included gym class, eating in the cafeteria, parties, riding on the school bus, speaking up in class, and talking to teachers or the principal. Once again, despite considerable discomfort, she did not avoid most of these situations.

In addition to her difficulties with school-related anxiety, the child was also observed to periodically engage in idiosyncratic speech. Specifically, she spoke in a style characteristic of much younger children, described by her parents as "baby talk". She spoke this way quite often,

although usually just among adults, and especially when the situation was aversive or boring. Her parents often scolded her for talking this way but stated they had minimal success in reducing its frequency.

At pretreatment assessment, her levels of crying, arguing, tantruming, and complaining about minor somatic distress or school were reported by her mother to be at a steady and frequent level, at least one or two episodes per day.

Functional Assessment

As part of the diagnostic evaluation, the child and parents completed the School Refusal Assessment Scale (SRAS; Kearney, 1993). The SRAS is a 24-item self report measure for children and their parents designed to identify maintaining variables and functional relationships associated with school refusal behavior. The SRAS has previously demonstrated adequate validity and reliability (Kearney, 1993). For this subject, scores from the mother, father, and child versions of the SRAS were combined into a composite set of scores that are illustrated in Figure 1. Because the variability of the scores is often limited, the scores are not meant to be interpreted in terms of their absolute magnitude; rather, the relative ranking of the factors is of greatest interest (Durand, 1990; Kearney, 1993). The assessment suggested that "attention getting/separation anxious" was the principle factor score, and hence, the primary function of the child's school refusal behavior. In accordance with the logic of prescriptive treatment, a treatment plan involving shaping and differential reinforcement was chosen.

Treatment Design And Procedure

Treatment was administered by the first author, a clinical psychology doctoral candidate, in the course of weekly sessions. During the first session, the practice of daily monitoring was discussed with the child and her mother. Monitoring was described as a necessary part of understanding the problems and how they might change over time. The child was instructed to write down in her diary any anxious or upsetting episodes as well as associated subjective units of distress (SUDS), a rating of fear on a scale ranging from 0 to 8. In addition, after the child had been excused from the session, her mother was asked to keep a diary recording all of the child's problem behaviors related to school refusal. The mother's monitoring was intended to be done outside of the child's awareness in order to minimize reactivity that was expected from the child. This formulation was derived from information from the SRAS, which suggested that mother's attention (e.g., overt monitoring) might be a reinforcer for attention seeking behavior and could thus increase the occurrence of target behaviors.

The second session was used to ensure that monitoring was going as instructed and to begin to organize the target behaviors into meaningful domains for a multiple-baseline intervention. By this session, the mother's monitoring was proceeding in a systematic and organized manner. The girl's diary, however, was poorly completed. She made only a few entries, and these were limited to rather trivial somatic complaints (e.g., "my thumb was itchy"). It was clear from the discrepancy in monitoring that the child's self-report would not be a reliable source of assessment data. Although the child diary was continued throughout the course of treatment for general clinical purposes, those data were not used to evaluate the intervention. The mother's data, on the other hand, were extensive and were thus organized into four general categories of problem behaviors: somatic complaints, anger and tantrums, tears, and other complaints. By the end of the second session, it was agreed that these four domains would be examined in a

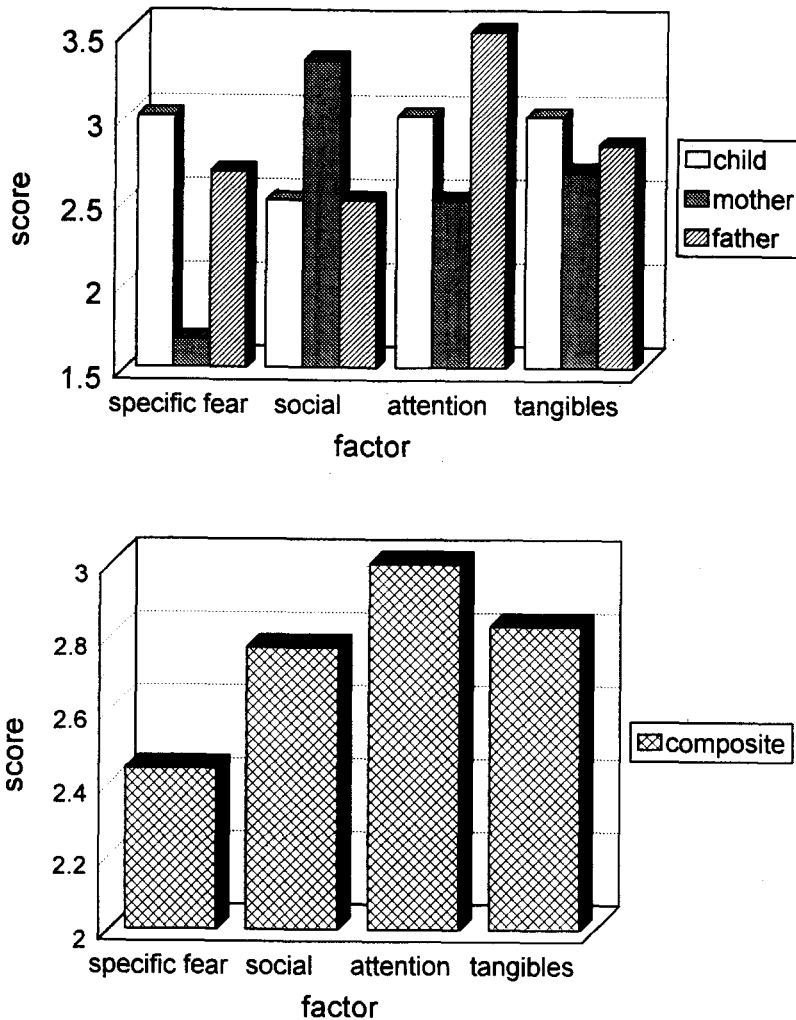


Figure 1. Individual and composite scores from the School Refusal Assessment Scale. Note: "specific fear" = specific fearfulness or general over-anxiousness; "social" = escape from aversive social stimuli; "attention" = attention-getting or separation anxious behavior; "tangibles" = tangible reinforcement.

multiple baseline across behaviors. Both the child and her mother were instructed to continue monitoring for another week before beginning any part of the intervention, thus providing a 2 week initial baseline.

The intervention was introduced in the third session. Consistent with the prescriptive approach for "attention seeking or separation anxious" behavior, differential reinforcement was suggested to minimize problem behaviors. The intervention was described as the consistent and vigilant delivery of parents' attention only to those child behaviors other than the one chosen for intervention for that week. That is, the target behavior would be completely ignored and placed on an extinction schedule. The child and her mother were told how certain problems can be made to persist simply by "poor communication" among family members. The purpose of

the intervention, it was explained, was to change these communication patterns and to see if that helped to reduce difficulties associated with school attendance. Parents were told not to "ignore their child", but simply to ignore her problem behaviors.

In addition, activities in the family were used as functional substitutes for the child's school refusal behavior. That is, other ways to get attention were set up so that the child would have efficient and appropriate methods to get the attention she desired from her parents. For example, she chose cooking with her mother as an activity she would like to do for herself and her family. Her mother was instructed to give her much praise for these behaviors and to have the father do the same. The goal was to create a reinforcement schedule for behaviors functionally equivalent to the target behaviors. The use of naturally occurring reinforcers over contrived praise or other types of reinforcers was selected because of more favorable implications for relapse prevention (cf. Ferster, 1967).

Session 3, then, began with the application of this intervention to the first domain of behavior, somatic complaints. Mother was told that any other problems behaviors were to be attended to as usual. This was continued through session 4, which involved fine-tuning of some problems using role play and discussion. For example, the child was asked to act out a challenging episode of school refusal behavior (e.g., complaining, seeking reassurance, saying "you don't care about me") in the session to allow the mother the opportunity to practice selectively ignoring these target behaviors. These role-plays allowed for identification of the more challenging behaviors (i.e., most difficult to ignore) and hence allowed the mother to be alert and more prepared for their occurrence at home. In addition, specific feedback was provided following these role-plays to reinforce the mother for proper administration and to point out where she had either failed to ignore or had over-generalized the intervention (i.e., ignored non-target behaviors). Finally, these role plays also assisted the child in understanding the nature of the contingency being established for home.

In sessions 5 and 6, anger and tantrums were targeted using the differential reinforcement intervention for 2 weeks, and then crying was targeted in the following two weeks (sessions 7 and 8). Other complaints were never targeted directly but were not observed following session 8 (see Figure 2).

Results

Results are summarized in Figure 2. Baseline monitoring demonstrated substantial school refusal behavior across all four domains. Although variability during baseline was moderate, the trend and level of these behaviors were rather stable and indicated a steady presentation of school refusal behavior before the intervention was introduced.

The somatic complaints targeted in the first phase of the intervention decreased to zero frequency after the first 3 days. During the last 2 of those same 3 days, the behaviors in the non-targeted domains were not observed. Thereafter, these non-targeted behaviors returned to their baseline level for the remainder of the first phase of treatment. Somatic complaints remained at a reduced level throughout treatment, occurring only once in the remaining 60 days. The second phase change was marked by immediate decrease in the target behavior. Anger and tantrum behavior returned at day 40 to day 45, but was not observed again. The remaining non-targeted behaviors initially decreased to minimal levels during this second phase of treatment, but returned to near baseline level at approximately day 35. The third phase change also showed immediate termination of the targeted behavior. All other target behaviors

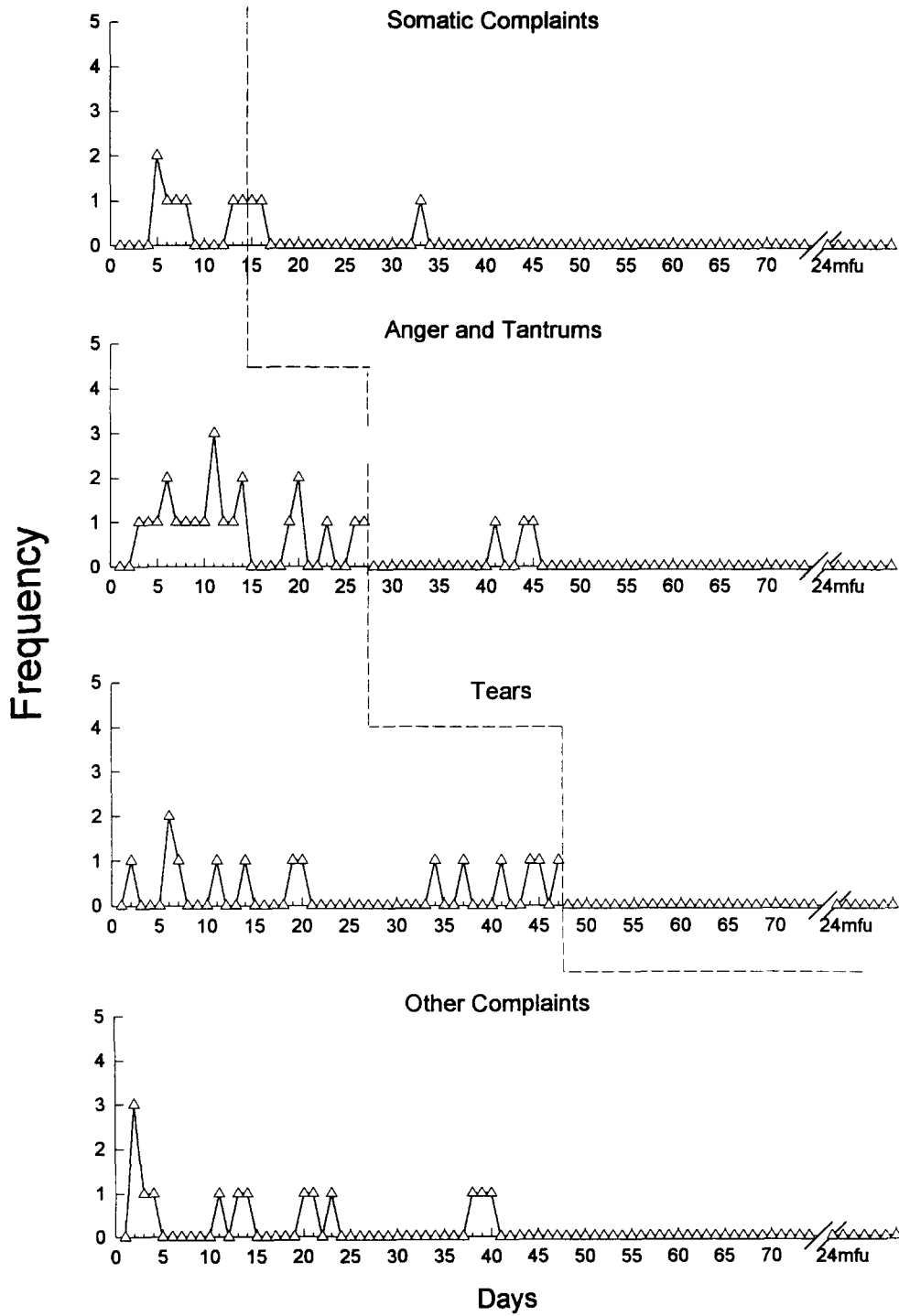


Figure 2. Multiple baseline across four target behaviors demonstrating frequency per day.

were also observed to have ceased or remained at zero frequency, and thus, no further intervention was introduced targeting other complaints.

These reductions in the frequency of school refusal behavior were maintained at a 2 week post-treatment assessment. Post-treatment independent diagnostic evaluation yielded diagnoses of Separation Anxiety Disorder, full remission and Social Phobia, full remission. At the 24 month follow-up assessment, no criteria were met for any diagnosis, and 1 week of monitoring demonstrated no occurrence of problem behaviors. In addition, retrospective reporting at the follow-up assessment indicated no notable recurrence of the target behaviors or diagnoses during the 2 year interval.

Discussion

The present study is informative in that it demonstrates a successful intervention while departing rather dramatically from the typical topographical approach of matching diagnosis to treatment. Indeed, the comprehensive treatment for separation anxiety disorder and social phobia would have involved other interventions focused on these specific diagnoses while only indirectly addressing the concomitant school refusal behavior. The data suggest that in cases of school refusal, a functional, prescriptive approach directed primarily at the child's school refusal behavior may be the most parsimonious strategy.

One notable pattern in the data concerns the potential role of rule governance in modifying children's behavior. That is, the therapist's having described each phase change and associated contingency shift to the child might well have facilitated the extinction of attention-seeking behavior. Indeed, in this particular case, the "extinction burst" one would typically expect in an extinction paradigm did not occur. The child did test the contingency during the first extinction phase, but not thereafter. One would not typically expect behavior to extinguish immediately in the absence of rule governance. Of course, such a hypothesis would have been better tested by only alerting the child to some phase changes and not others. Thereby, one could compare the rate of extinction for rule-governed vs purely operant conditioning trials. Nevertheless, the pattern suggests the importance of stating contingency changes clearly to the child in order to minimize extinction bursts or other potential problems.

One particular problem in assessing the integrity of the intervention involved inadequate monitoring of positive behavior. The prescriptive approach hypothesizes that a functional substitute for the problem behavior is an integral part of the intervention. However, there was no parent monitoring of the child's positive behaviors (e.g., cooking) to support the hypothesis that increases in functionally equivalent positive behaviors will assist with decreases in the target behaviors. One would suspect that if this positive behavior showed an increase corresponding with the decrease of negative attention seeking, then the inclusion of a functional substitute was a key ingredient in the treatment. Otherwise, one might argue that extinction alone was sufficient for treatment. Although the frequency of positive attention seeking was noted to increase, such data were not gathered systematically. Future studies should evaluate the necessity of this positive behavior component of treatment.

Another concern involved the reliability of the diary measurements. The use of multiple raters as a reliability check is not often feasible in the treatment of child cases. Because much of the behavior involved the mother-child dyad, it was especially difficult to identify a second adult who would be able to accurately monitor the child's behavior.

Another pattern worth noting involved the apparent lack of treatment specificity within

phases. That is, treatment seemed to generalize a bit to domains not yet targeted by the differential reinforcement intervention, particularly to the last target. This may reflect a number of different phenomena. First, it is possible that some exogenous or non-specific variable had its effect on all problem behaviors. For example, not wanting to come to treatment any more may have made the child work harder to control her behavior. What seemed to be more probable, however, was that the final category represented non-specific behaviors. For example, over the course of treatment, it became clear that "other complaints" almost never occurred in the absence of "anger and tantrums". Thus, the behavior change noted in the fourth domain may not represent generalization of the intervention. This highlights the difficulty of establishing well-defined and independent categories within which to intervene in a multiple-baseline design, a process often conducted at the beginning of treatment, when one is least familiar with the child.

One final comment involves the assessment instrument itself. The SRAS used in this study was a preliminary version of a measure that has recently been updated. Psychometric support for the SRAS is somewhat limited, and its validity stems mainly from earlier treatment-outcome work (Kearney, 1995a). Future study investigating a functional treatment approach for school refusal behavior should include and evaluate the School Refusal Assessment Scale—Revised (Kearney, 1995b), a measure conceptually identical to the SRAS that has been revised to improve its administration and psychometrics (see Kearney, 1995b).

In general, school refusal behavior reflects an enormous complexity and diversity, often challenging to the clinician. An increased understanding of school refusal will most likely result from the continued dissemination of expedient approaches to functional analysis, rather than those approaches that employ non-specific strategies or that are limited to diagnosis-to-treatment matching. The prescriptive approach (Kearney & Silverman, 1990; Burke & Silverman, 1987), while perhaps not providing a definitive division or classification of the phenomenon, appears to have preliminary support. Unquestionably, the value of research that continues to evaluate and refine the meaningful dimensions of school refusal behavior and their appropriate treatments cannot be overstated.

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